Digital Signal Processing

Introduction

Prof. Aldebaro Klautau, Ph.D.
Federal University of Pará (UFPA)

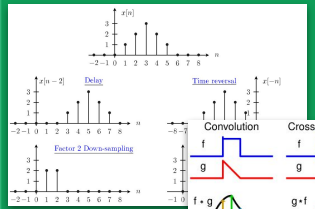
www.lasse.ufpa.br & www.ppgee.ufpa.br

Course topics

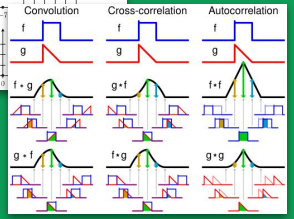
Main book: https://ai6g.org/books/dsp/ak_dsp_book.html

Chapter 1: Signals

- Types
- Manipulation
- Quantization
- Sampling
- Correlation



https://signalprocessing-engineer.netlify.app/signal-processing-fundamentals/signal-processing-basics/signal-manipulations/

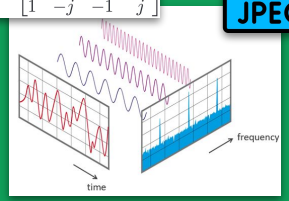


https://en.wikipedia.org/wiki/correlation

Chapter 2: Transforms

- DCT
- DFT
- Fourier
- Laplace
- Z

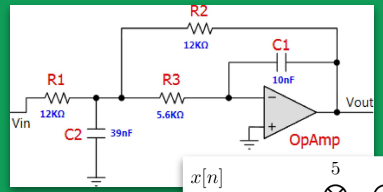
$$A_{DFT}^{-1} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

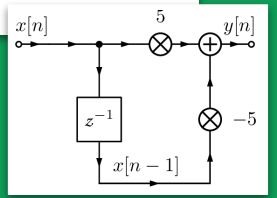


https://www.ohio.edu/mechanical-biomedical-biomedical-engineering/sites/default/files/sites/mechanical-biomedical-biomedical/pdf/ME6700.pdf

Chapter 3: Systems

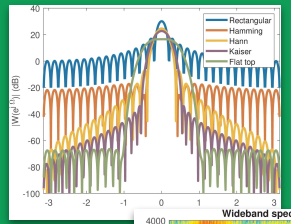
- Filter Design
- Reconstruction
- Bilinear

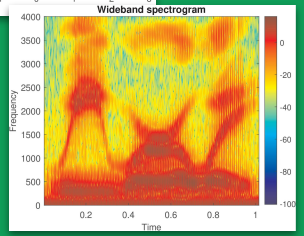




Chapter 4: Spectral Estimation

- Windows
- PSD
- Spectrogram





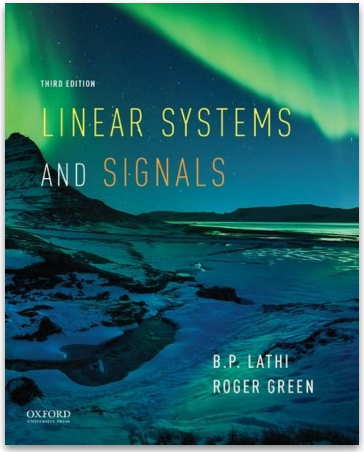
Course sources

Other books:

- Oppenheim

A. V. Oppenheim, R. W. Schafer. *Processamento em tempo discreto de sinais*, 3a Ed., Pearson, 2013.

- Lathi



B. P. Lathi, R. Green. *Linear Systems and Signals*, 3rd Ed., Oxford University Press, 2018.

Important links:

- https://github.com/lasseufpa/dsp2026
- https://nextcloud.lasseufpa.org/s/goDnjjE4HAF8rxi

Evaluation Process

Activity	Score
Weekly Exercises	5%
Exams	65%
Guided Projects	10%
Final Project	20%